

Chapter N was amended by [Local Law 77 of 2023](#). This law has an effective date of June 10, 2023.

APPENDIX N

ASSISTIVE LISTENING SYSTEMS PERFORMANCE STANDARDS

SECTION BC N101 GENERAL

N101.1 Scope. The provisions of this appendix shall control the performance standards of assistive listening systems (ALS).

SECTION BC N102 INDUCTION LOOP SYSTEM

N102.1 Induction loop system specifications.

1. Where induction-loop systems are provided for hearing aid purposes, the system performance shall be in accordance with IEC 60118-4:2014.
2. The methods of measuring and specifying the performance of system components shall be in accordance with IEC 62489-1:2010, as specified in IEC 60118-4:2014.
3. All electrical wiring installations shall be in accordance with the *New York City Electrical Code*.

N102.2 Capital projects. City funded projects shall provide induction-loop systems pursuant to Section 224.3 of the New York City Charter.

SECTION BC N103 INFRA-RED SYSTEM

N103.1 Infra-red system specifications.

1. If the ambient infra-red light (generally caused by other infra-red light-generating devices or direct sunlight) produces interference, it is recommended that, unless the interfering infra-red light can be reduced, infra-red systems not be considered.
2. Given an input signal to the transmitter at the levels specified in Item 6 over a minimum frequency range of 200 Hz to 8,000 Hz, the measured frequency response of the output signal generated by the receiver and transducer over that range shall not vary by more than +/- 5 decibels from the value at 1,000 Hz. In actual usage, graphic equalization of the output signal is allowed.
3. Given an input signal to the transmitter at the levels specified in Item 6, the minimum signal-to-noise ratio of the output generated by the receiver and transducer shall be 35 decibels at all frequencies specified in Item 2.
4. Given an input signal of 1,000 Hz to the transmitter at a level sufficient to produce the minimum signal-to-noise ratio specified in Item 3, the harmonic distortion produced by the receiver and transducer shall not be more than 5 percent.
5. Given a 1,000 Hz sinusoidal input signal at the levels specified in Item 6, the following are the possible transducers:
 - 5.1. If using a neckloop as an output transducer, generate a magnetic field strength of at least 150 mA/meter. The peak field strength shall not exceed 600 mA/meter. Measurements shall be made at the geometric center of the plane of the neckloop.

- 5.2. If using a silhouette as an output transducer, generate a magnetic field strength of at least 50 mA/meter (the peak field strength shall not exceed 200 mA/meter). Measurements shall be made at a distance of 10 centimeters from the silhouette.
- 5.3. If using a miniature earphone as an output transducer, generate a sound pressure level of at least 80 decibels. The maximum high-frequency average sound pressure level shall not exceed 130 decibels.
- 5.4. If a direct input cable is used between the output of the receiver and the input of a hearing aid boot or shoe, generate a minimum voltage range of 2 mV to 800 mV and match a boot/shoe input impedance of 50 ohms or greater. The plugs at either end shall be molded 2-pin, 3-pin and/or 3.5-millimeter mini-plug (stereo or mono).
6. The system must have inputs capable of accepting signals at line level and microphone level and must be capable of interfacing with existing public address systems or stand-alone.
7. The minimum light level of the transmitted infra-red carrier signal must be sufficient so that a receiver can produce the specifications in Items 2, 3, 4 and 5 at any point within the user area.
8. Any stand-alone devices powered by 110-120 volt AC used as part of the system must be tested by an approved laboratory.
9. All electrical wiring installations shall be in accordance with the *New York City Electrical Code*.

**SECTION BC N104
FM SYSTEM**

N104.1 FM system specifications.

1. If an ambient FM signal (generally caused by other agencies on the same frequency or major power supplies) produces interference, it is recommended that, unless the frequency can be changed, an FM system not be considered.
2. Given an input signal to the transmitter at the levels specified in Item 6 over a minimum frequency range of 100 Hz to 8,000 Hz, the measured frequency response of the output signal generated by the receiver and a transducer over that range shall not vary by more than +/- 5 decibels from the value at 1,000 Hz. In actual usage, graphic equalization of the output signal is allowed.
3. Given an input signal to the transmitter at the levels specified in Item 6, the minimum signal-to-noise ratio of the output generated by the receiver and transducer shall be 35 decibels at all frequencies specified in Item 2.
4. Given an input signal to the transmitter at 1,000 Hz. at a level sufficient to produce the minimum signal-to-noise ratio specified in Item 3, the total harmonic distortion between the input signal and the program signal produced by the receiver and transducer shall not be more than 10 percent.
5. Given a 1,000 Hz. sinusoidal input signal at the levels specified in Item 6, the following are among the possible transducers:
 - 5.1. If using a neckloop as an output transducer, generate a magnetic field strength of at least 150 mA/meter. The peak field strength shall not exceed 600 mA/meter. Measurements shall be made at the geometric center of the plane of the neckloop.
 - 5.2. If using a silhouette as an output transducer, generate a magnetic field strength of at least 50 mA/meter. The peak field strength shall not exceed 200 mA/meter. Measurements shall be made at a distance of 10 centimeters from the silhouette.
 - 5.3. If using a miniature earphone as an output transducer, generate a sound pressure level of at least 80 decibels. The maximum high-frequency average sound pressure level shall not exceed 130 decibels.
 - 5.4. If a direct input cable is used between the output of the receiver and the input of a hearing aid boot or shoe, generate a minimum voltage range of 2 mV to 800 mV and match a boot/shoe input impedance of 50 ohms or greater. The plugs at either end shall be molded 2-pin, 3-pin and/or 3.5-millimeter mini-plug (stereo or mono).
6. The system must have inputs capable of accepting signals at line level and microphone level and must be capable of interfacing with existing public address systems or stand-alone.

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7. The minimum sensitivity of the receiver shall be at least 2 mV at 12 decibel SINAD. The maximum RF signal generated by the transmitter shall not exceed 8,000 mV/meter at 30 meters.
8. Any stand-alone devices powered by 110-120 volt AC used as part of the system must be tested by an approved laboratory.
9. All electrical wiring installations shall be in accordance with the *New York City Electrical Code*.
10. The frequencies used by the transmitter should be in compliance with applicable FCC rules.